

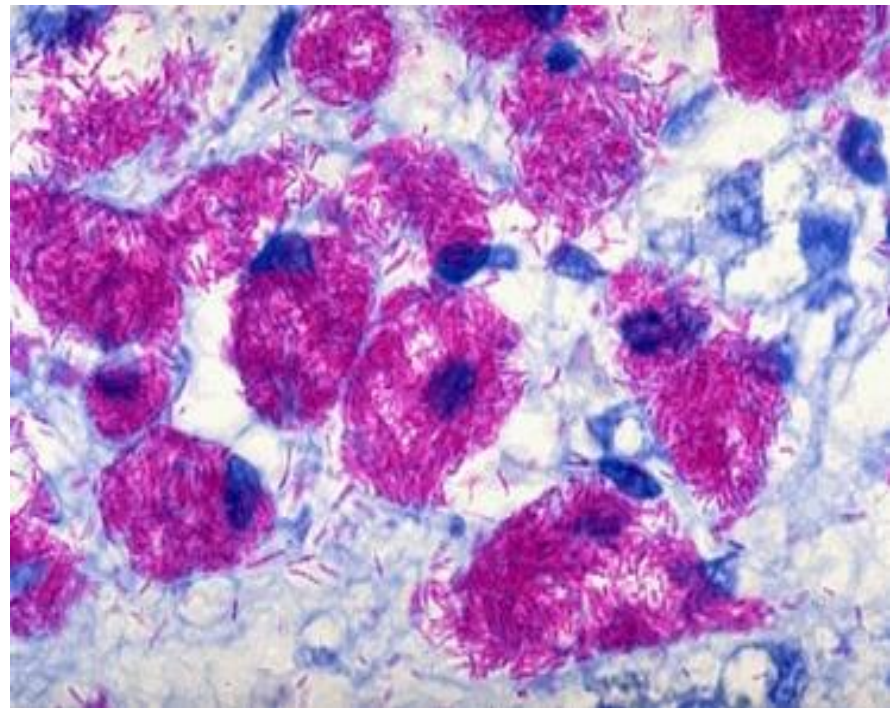
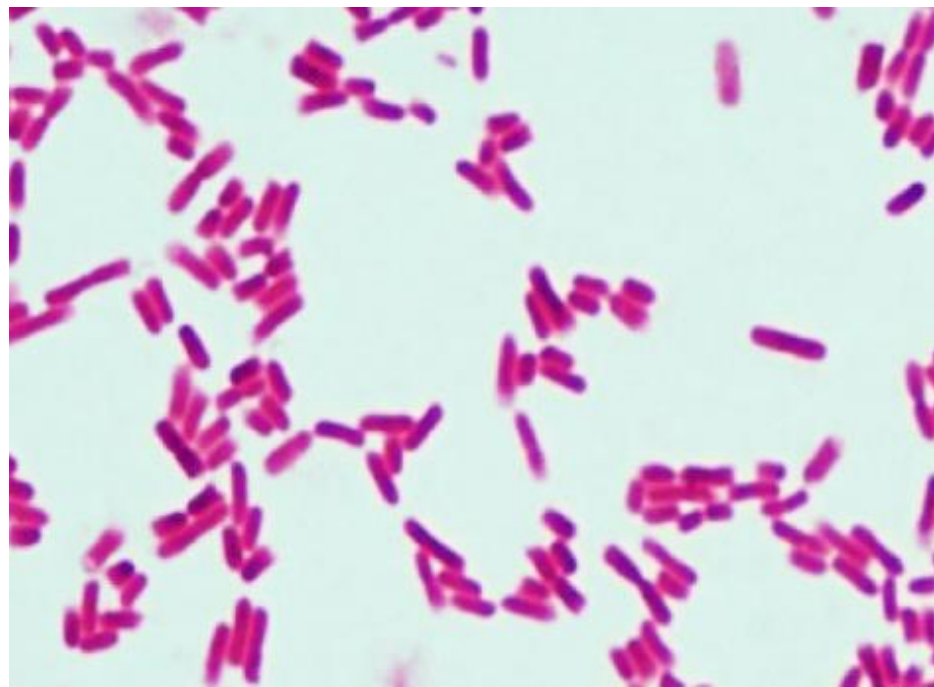
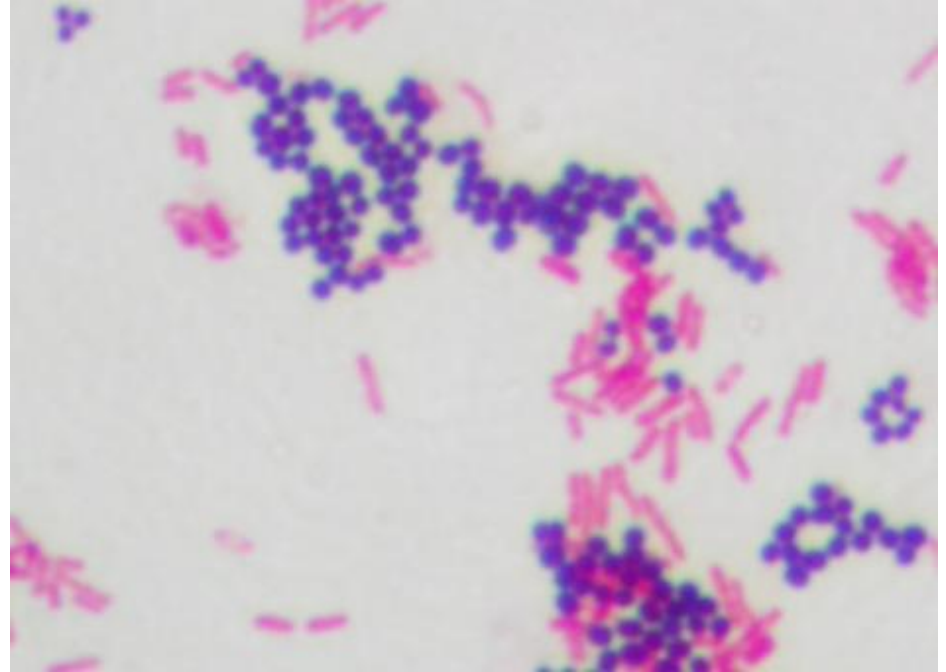
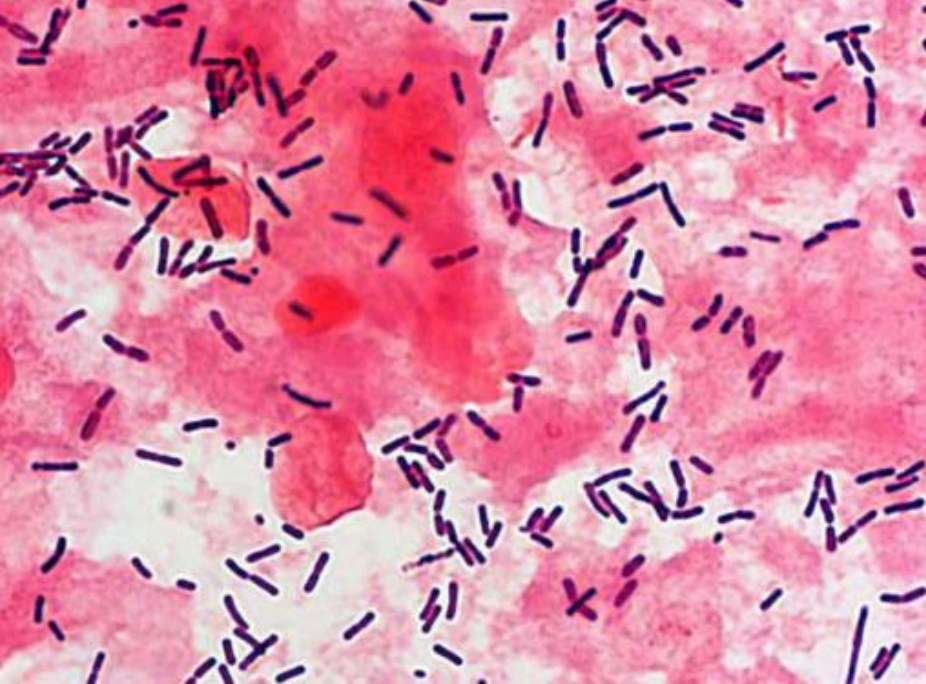
Histopathology and cytopathology
of bacterial disease , viral and some
parasitic diseases

TECHNIQUES FOR DIAGNOSING INFECTIOUS AGENTS

- Many infectious agents are best visualized by special stains that identify organisms on the basis of their cell wall characteristics including Gram, acid-fast, silver, mucicarmine, and Giemsa stains

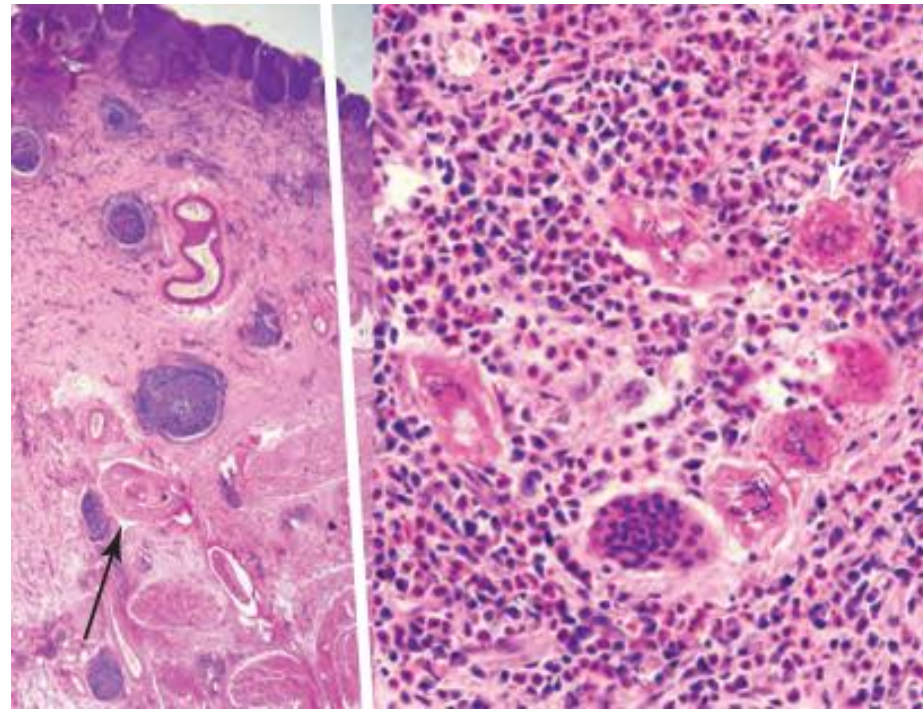
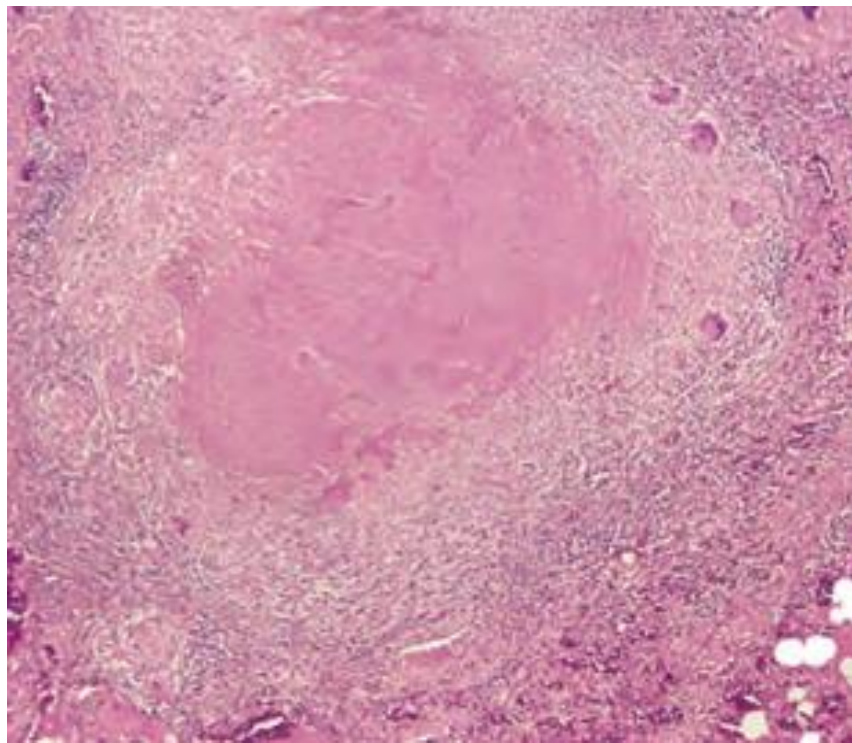
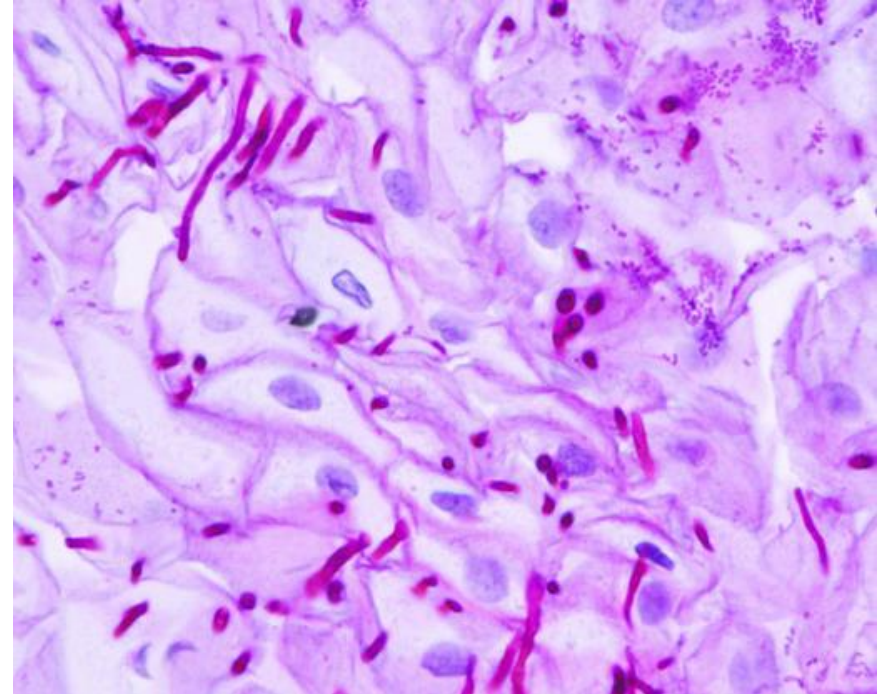
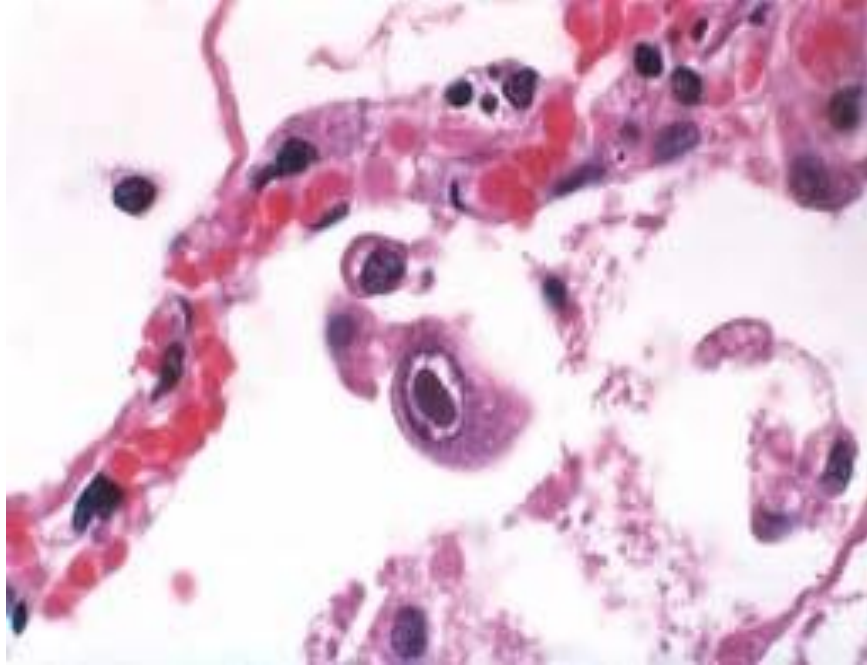
Cont

- Gram stain : most bacteria
- Acid fast stain: mycobacteria
- Silver stain: fungi, pneumocysts, legionellae
- Periodic acid-Schiff: fungi, amebae
- Mucicarmine: cryptococci
- Giemsa: campylobacter., leishmania., malaria



Cont.

- Some infectious agents can be directly identified in H+E-stained sections (e.g., CMV or herpesvirus inclusion bodies; bacterial clumps, which usually stain blue)
- *Candida* and *Mucor* among the fungi;
- most protozoans; and all helminths.



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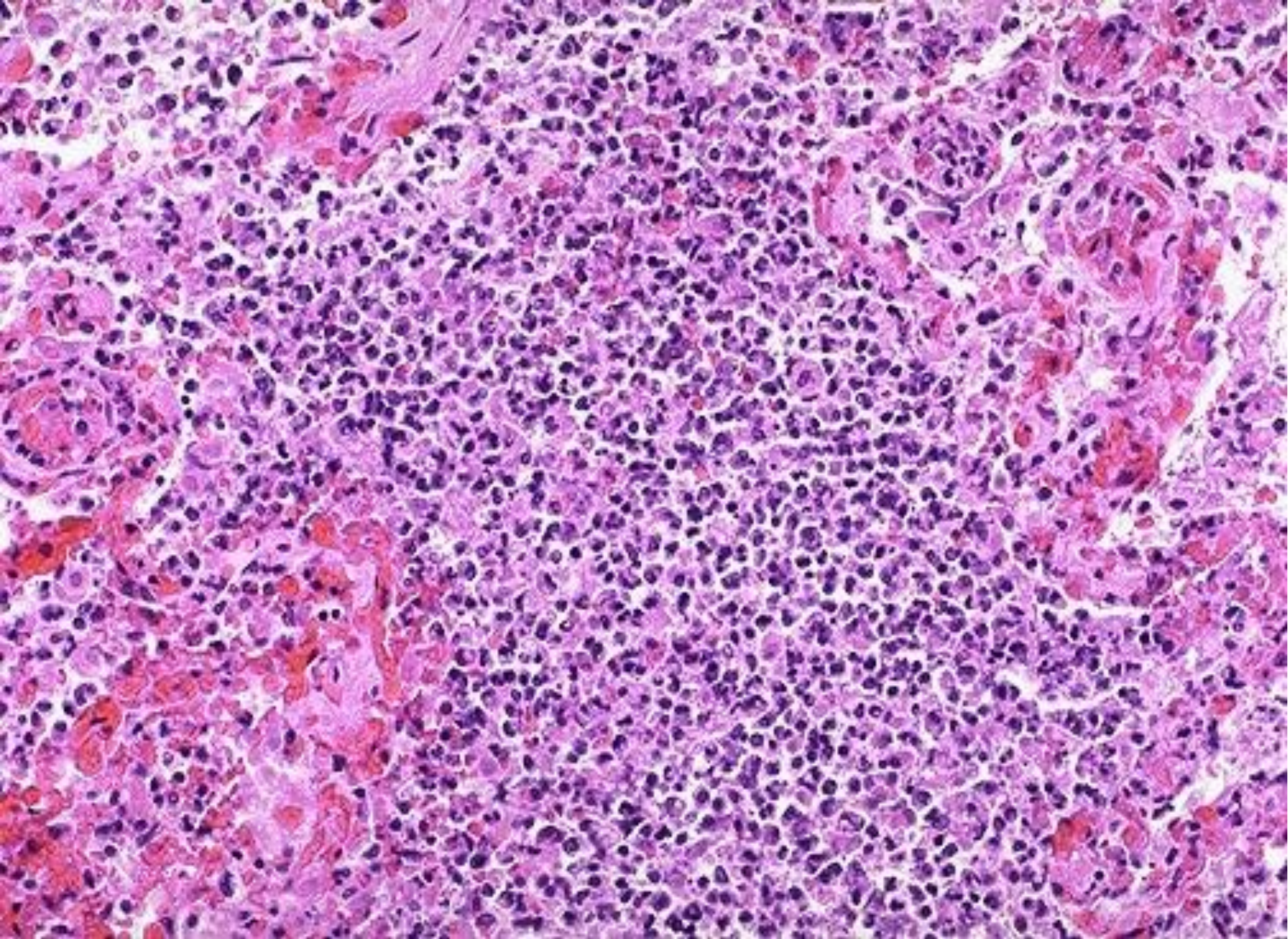
- Nucleic acid amplification tests such as polymerase chain reaction (PCR) are now used for diagnosis of gonorrhea, Chlamydia infection, tuberculosis, and herpes encephalitis
- Culture and antibody stain

Patterns of Inflammatory Responses to Infection

- Although infectious microorganisms themselves are wildly diverse, the infected host actually has relatively few ways that it can respond.
- Thus, at the microscopic level many pathogens evoke similar reaction patterns
- Nevertheless, the different patterns of response do suggest particular classes of causative organisms .

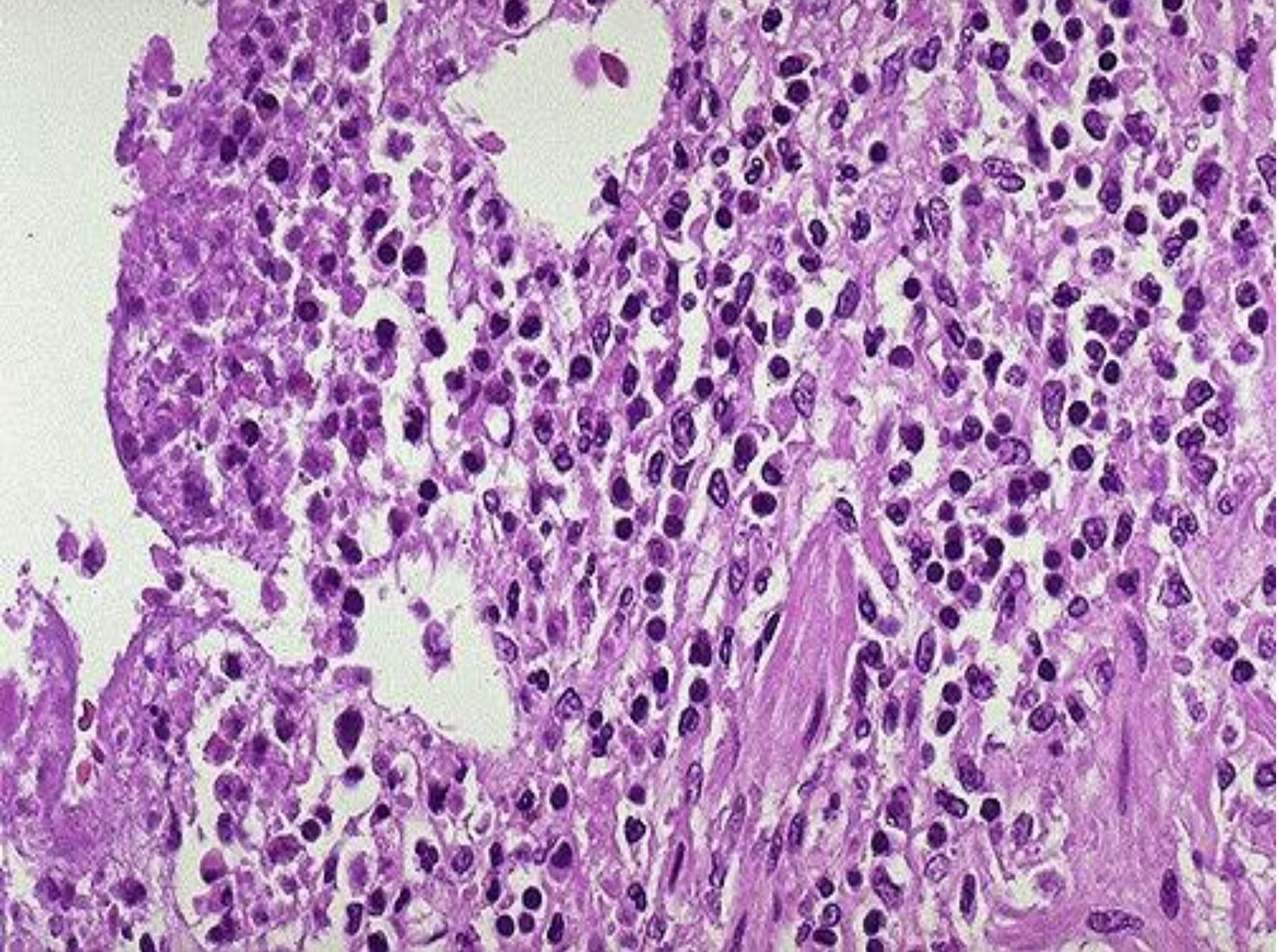
Suppurative Inflammation

- This pattern is the reaction to acute tissue damage , characterized by increased vascular permeability and leukocytic exudates, predominantly neutrophils.
- In many cases, this is a response to ***extracellular bacteria***. The neutrophils are typically recruited to the site of infection by chemoattractants released from "pyogenic" organisms and from host cells



Mononuclear and Granulomatous Inflammation

- Diffuse mononuclear interstitial infiltrates are a common feature of all chronic inflammatory processes, but when they develop acutely, they often are a response to viruses, intracellular bacteria, or intracellular parasites.
- In addition, spirochetes and helminthes provoke chronic inflammatory responses.

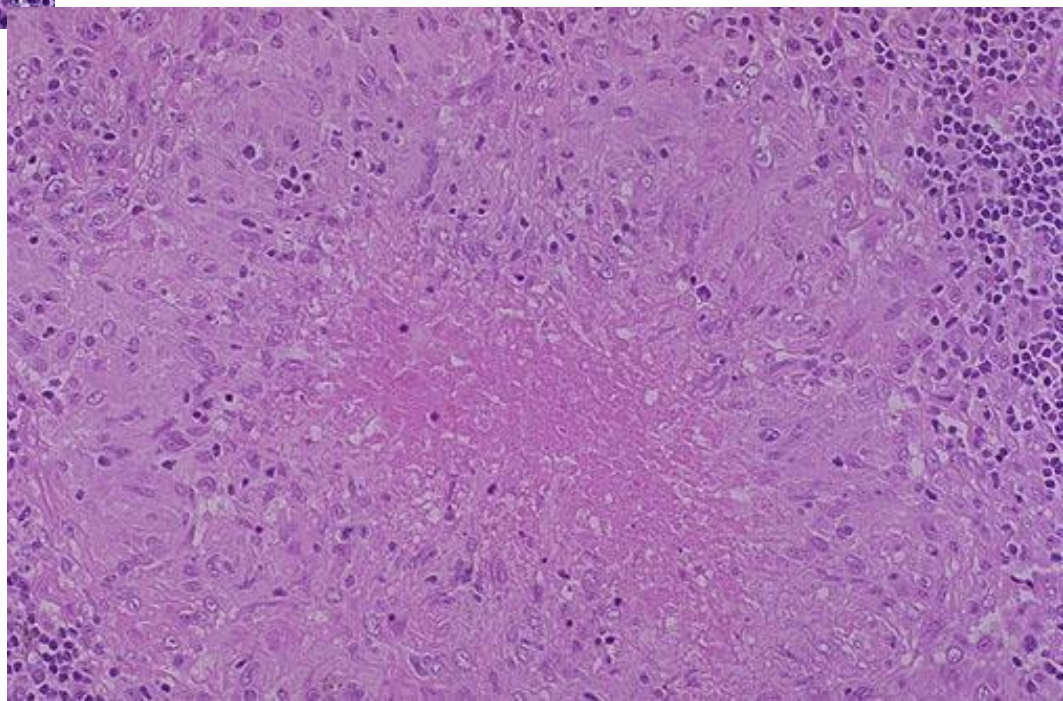
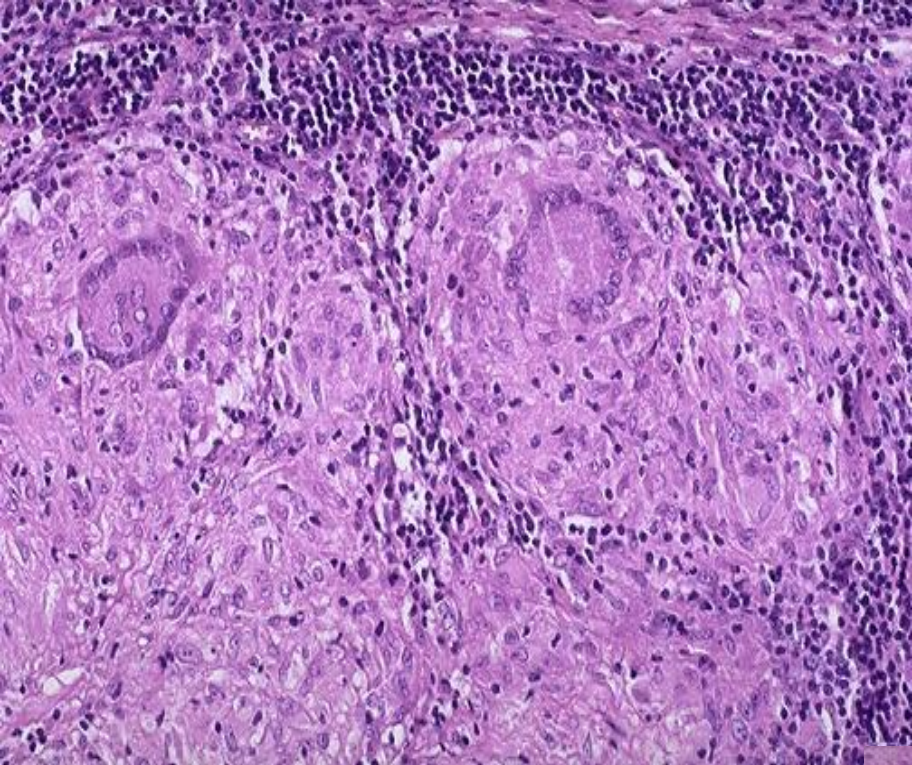


Cont.

- **Granulomatous inflammation** is a distinctive form of mononuclear inflammation usually evoked by infectious agents that resist eradication (e.g., *M. tuberculosis*, *Histoplasma capsulatum*, schistosome eggs) but are nevertheless capable of stimulating strong T cell-mediated immunity.

Cont.

- Granulomatous inflammation is characterized by accumulation of activated macrophages called "epithelioid" cells, which may fuse to form giant cells. In some cases, there is a central area of caseous necrosistb



Cytopathic-Cytoproliferative Response

- These reactions are usually produced by viruses and are characterized by sparse inflammation and either cell death (cytopathic response) or proliferation (cytoproliferative response).

Cont.

- Some viruses replicate within cells and make viral aggregates that are visible as inclusion bodies (e.g., CMV, HSV, HBV) or induce cells to fuse and form polykaryons (e.g., measles, herpesviruses)

Cont.

- Focal cell damage may cause epithelial cells to become discohesive and form blisters (e. herpes simplex virus).
- Viruses can also cause epithelial cells to proliferate and take unusual forms (e.g., venereal warts caused by HPV or the umbilicated papules of molluscum contagiosum caused by poxviruses)
- Finally, viruses can cause dysplastic changes and cancers in epithelial cells and lymphocytes



Necrotizing Response

- Some organisms produce potent toxins that cause such rapid and severe necrosis that tissue damage is the dominant feature (e.g., *C. perfringens* –gas gangrene).



Chronic Inflammation and Scarring

- The final common pathway of many infections is chronic inflammation, which can lead either to complete healing or to extensive scarring.
- For example, chronic HBV infection, Schistosoma eggs in liver or the constrictive fibrous pericarditis in tuberculosis

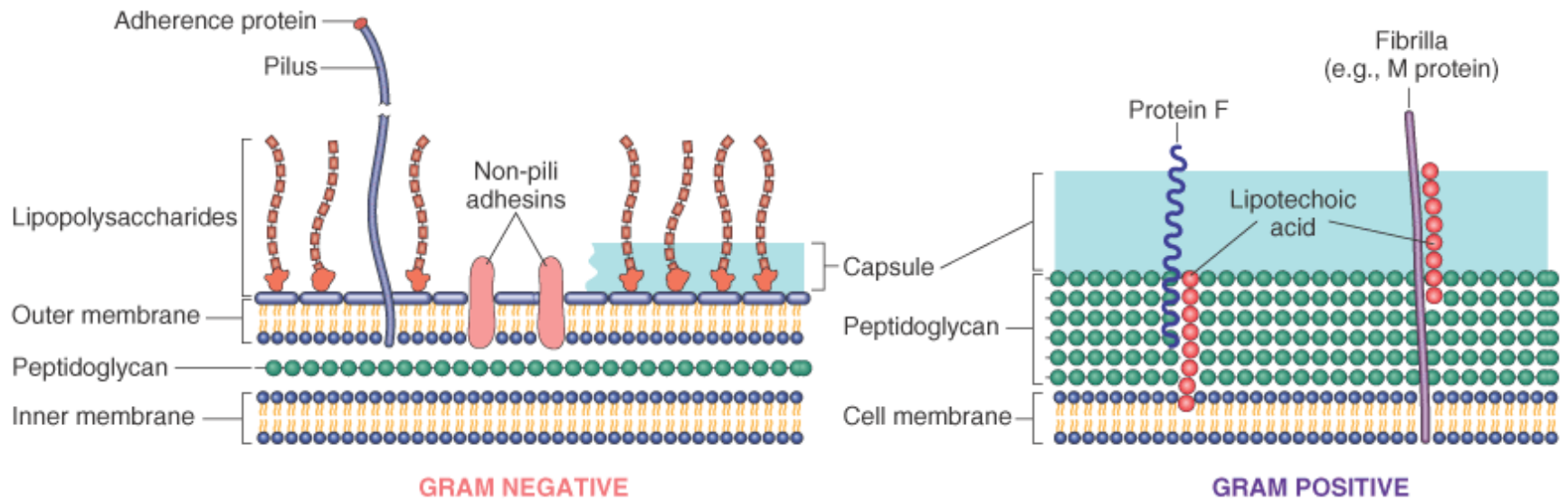
Bacterial infections

- Bacterial infections are common causes of disease.
- Bacterial cells have a cell membrane but lack membrane-bound nuclei and other membrane-enclosed organelles.
- bound by a cell wall usually consisting of *peptidoglycan*, a polymer made up of a mixture of sugars and amino acids.

Cont.

- Bacterial cell walls generally occur in one of two varieties: a thick wall surrounding the cell membrane that retains crystal violet stain (*Gram-positive*) or
- A thin cell wall sandwiched between two phospholipid bilayer membranes (these do not retain crystal violet stain and are thus *Gram-negative*)).

Cont



Cont.

- Bacteria are classified by Gram staining (positive or negative), shape (e.g., spherical ones are *cocci*; rod-shaped ones are *bacilli*), and form of respiration (aerobic or anaerobic)

Gram positive bacteria

- Common gram-positive pathogens include Staphylococcus, Streptococcus, and Enterococcus, each of which causes many types of infections.

Cont.

- Diphtheria, listeriosis, anthrax, and nocardia are less common infections also caused by gram-positive rods

Staphylococcal Infections

- *S. aureus* are pyogenic gram-positive cocci that form clusters resembling bunches of grapes.
- *S. aureus* causes a myriad of skin lesions (boils, carbuncles, impetigo, and scalded-skin syndrome) as well as abscesses, sepsis, osteomyelitis, pneumonia, endocarditis, food poisoning, and toxic shock syndrome.
- *S. saprophyticus* is a common cause of urinary tract infections in young women.

Cont.

- Coagulase negative staphylococci, such as *S. epidermidis*, cause opportunistic infections in catheterized patients, patients with prosthetic cardiac valves, and drug addicts.

Pathogenesis

- *S. aureus* produces a multitude of virulence factors, which include surface proteins involved in adherence and evasion of the host immune response, enzymes that degrade host structures, secreted toxins that damage host cells, and proteins that cause antibiotic resistance.

Cont.

- Staphylococci infecting prosthetic valves and catheters have a polysaccharide capsule that allows them to attach to artificial materials and resist host cell phagocytosis.

Cont.

- Staphylococci also have on their surface protein A, which binds the Fc portion of immunoglobulins, allowing the organism to escape antibody-mediated killing.
- *S. aureus* produces multiple membrane damaging (hemolytic) toxins.

Cont.

- Super antigens produced by *S. aureus* cause food poisoning and toxic shock syndrome.
- Toxic shock syndrome came to public attention because of its association with the use of hyper absorbent tampons, which became colonized with *S. aureus* during use.
- It is now clear that toxic shock syndrome can be caused by growth of *S. aureus* at many sites, most commonly the vagina and infected surgical sites.

Cont.

- This syndrome is characterized by hypotension (shock), renal failure, coagulopathy, liver disease, respiratory distress, a generalized erythematous rash, and soft tissue necrosis at the site of infection.
- If not promptly treated, it can be fatal. Toxic shock syndrome can also be caused by *Streptococcus pyogenes*.

Consequences of staphylococcal infection

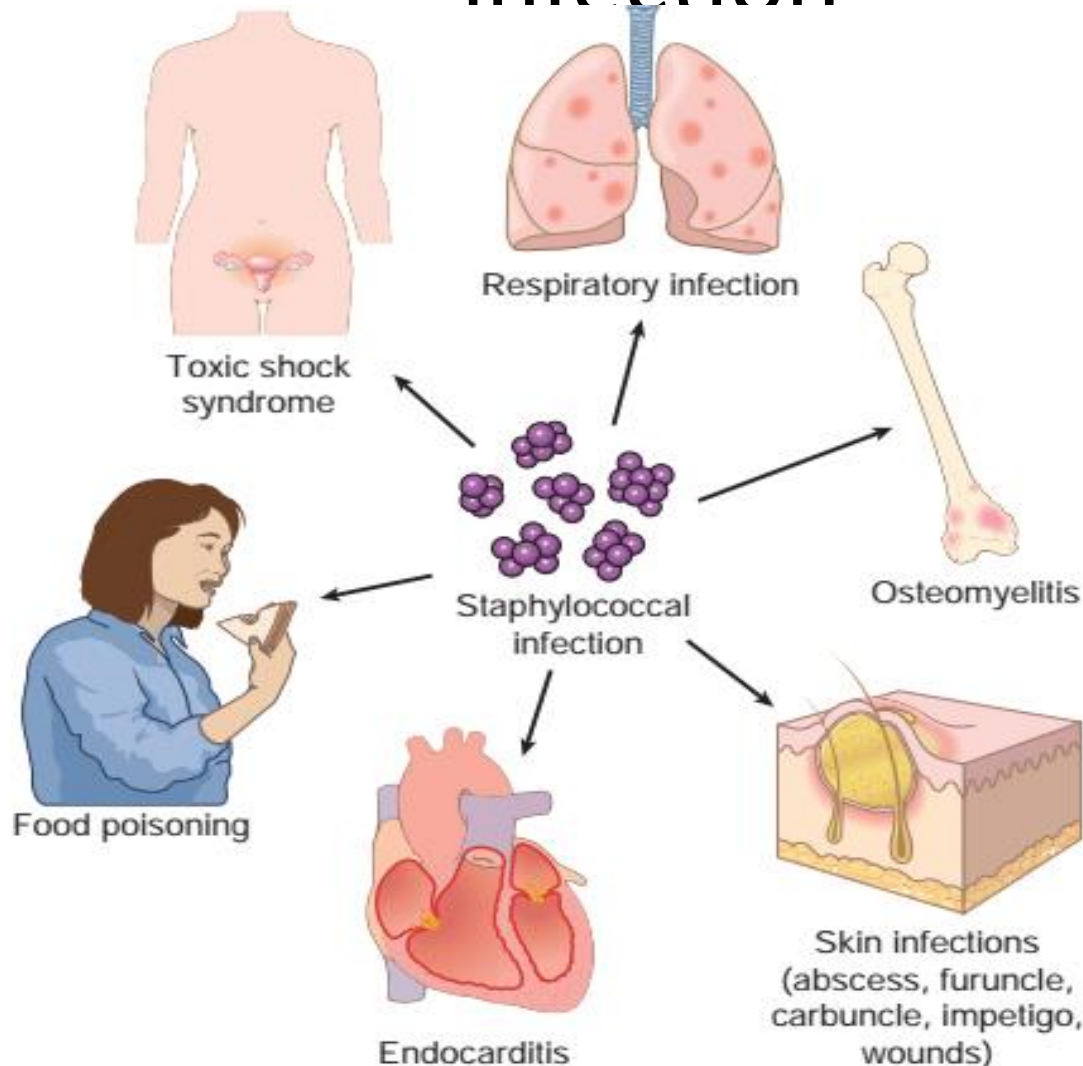


Figure 8-15 The many consequences of staphylococcal infection.

Microscopy

- Whether the lesion is located in the skin, lungs, bones, or heart valves, *S. aureus* causes pyogenic inflammation that is distinctive for its local destruction of host tissues.

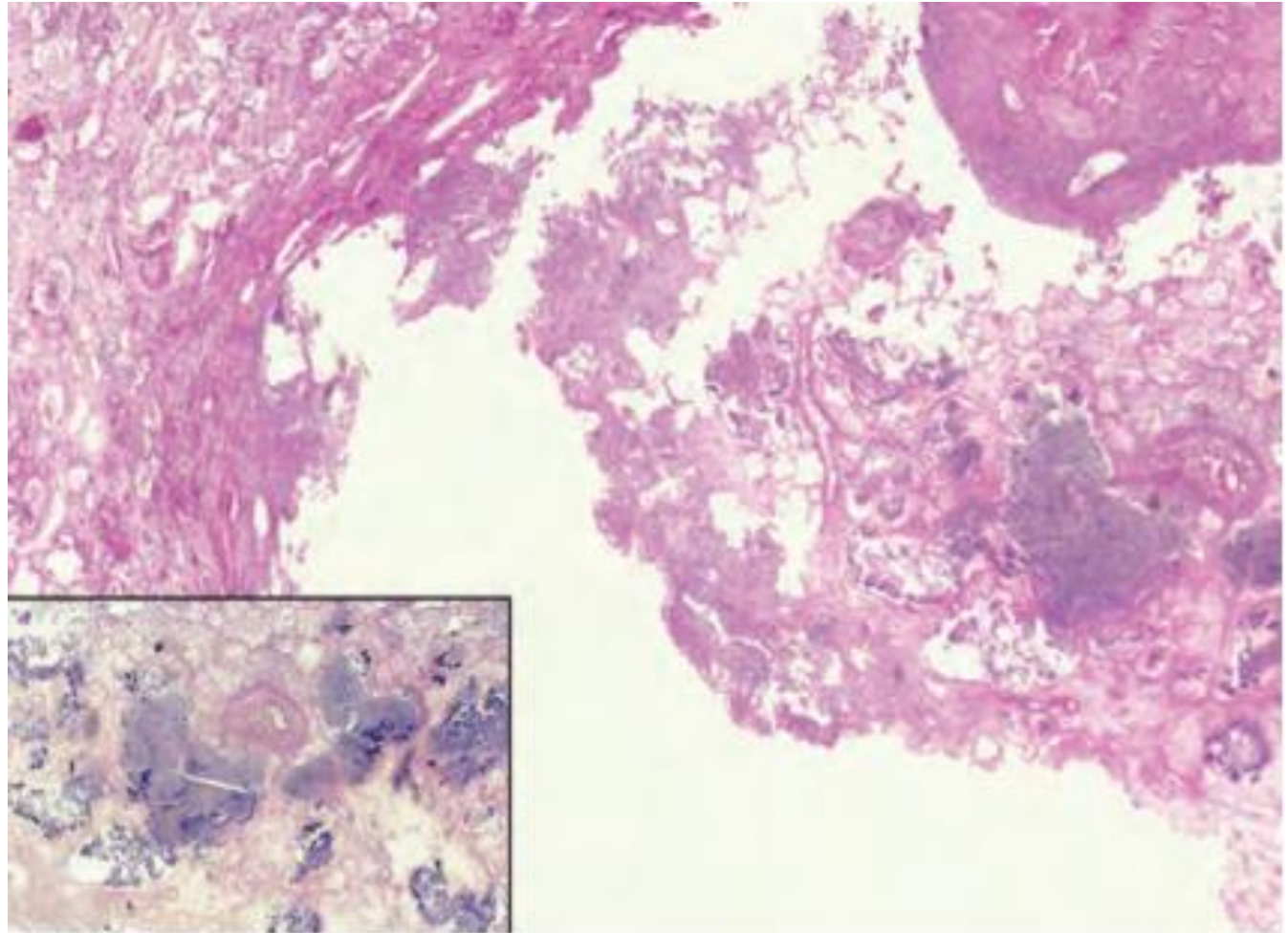


Figure 8-16 Staphylococcal abscess of the lung with extensive neutrophilic

streptococcal and Enterococcus Infections

- Streptococci cause suppurative infections of the skin, oropharynx, lungs, and heart valves.
- They are also responsible for a number of post-infectious syndromes
 - rheumatic fever
 - immune complex glomerulonephritis

Cont

- These bacteria are gram-positive cocci that grow in pairs or chains.
- *S. pyogenes* (group A) causes pharyngitis, scarlet fever, erysipelas, impetigo, rheumatic fever, toxic shock syndrome, and glomerulonephritis.

Cont.

- *S. agalactiae* (group B) colonizes the female genital tract and causes sepsis and meningitis in neonates and chorioamnionitis in pregnancy.
- *S. pneumoniae* is a common cause of community-acquired pneumonia in older adults and meningitis in children and adults.
- The viridans-group streptococci found in normal oral flora that are a common cause of endocarditis.
- *S. mutans* is the major cause of dental caries.

Pathogenesis

- The different species of streptococci produce many virulence factors and toxins. *S. pyogenes*, *S. agalactiae*, and *S. pneumoniae* have capsules that resist phagocytosis.
- *S. pyogenes* also expresses M protein, a surface protein that prevents bacteria from being phagocytosed, and a complement C5a peptidase that degrades this chemotactic peptide.

Cont.

- Virulent *S. pyogenes* have been referred to as flesh-eating bacteria because they cause a rapidly progressive necrotizing fasciitis.
- *S. mutans* produces caries by metabolizing sucrose to lactic acid which causes demineralization of tooth enamel
- Enterococci are low-virulence bacteria, although they do have an antiphagocytic capsule

Microscopy

- Streptococcal infections are characterized by diffuse interstitial neutrophilic infiltrates with minimal destruction of host tissues.
- The skin lesions caused by streptococci resemble those of staphylococci.

Cont

- Erysipelas is caused by exotoxins from superficial infection with *S. pyogenes*.
- The rash has a sharp, well-demarcated, serpiginous border and may form a “butterfly” distribution on the face.

Cont.

- On Histologic examination there is a diffuse, edematous, neutrophilic inflammatory reaction in the dermis and epidermis extending into the subcutaneous tissues.



Figure 8-17 Streptococcal erysipelas.

Diphtheria

- Diphtheria is caused by *Corynebacterium diphtheriae*, a slender gram-positive rod with clubbed ends that spreads from person to person in respiratory droplets or skin exudates.
- Cutaneous diphtheria causes chronic ulcers with a dirty gray membrane, but does not cause systemic damage.

Cont

- Respiratory diphtheria causes pharyngeal or, less often, nasal or laryngeal infection.
- There is toxin mediated formation of a gray pharyngeal membrane, and damage to the heart, nerves, and other organs.
- Release of exotoxin causes necrosis of the epithelium, accompanied by an out pouring of a dense fibrinosuppurative exudate.

Cont.

- The coagulation of this exudate on the ulcerated necrotic surface creates a tough, dirty gray to black, superficial membrane, sometimes called **pseudo-membrane** because it is not formed by viable tissue.
- There is an intense neutrophilic infiltration in the underlying tissues with marked vascular congestion, interstitial edema, and fibrin exudation.

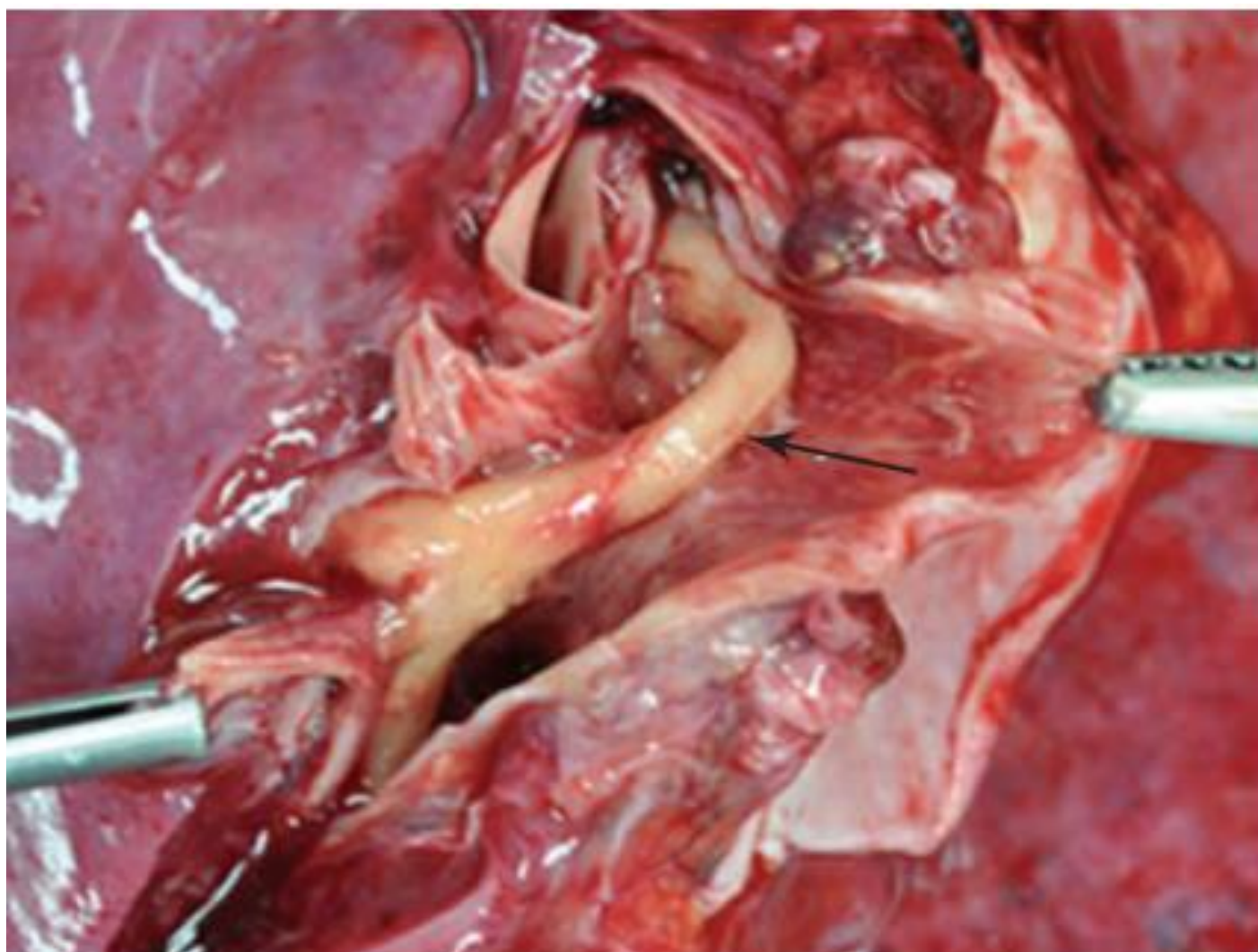


Figure 8-18 Membrane of diphtheria (*arrow*) lying within a transverse bronchus. (Courtesy Dr. Robin A. Cooke, Department of Anatomical Pathology, Princess Alexandra Hospital, Brisbane, Australia.)

Listeriosis

- *Listeria monocytogenes* is a gram-positive bacillus that causes severe food-borne infections in vulnerable hosts.
- Out breaks of *L. monocytogenes* infection have been linked to contaminated dairy products, chicken, and hot dogs.
- Pregnant women, neonates, older adults, and immunosuppressed persons are particularly susceptible to severe *L. Monocytogenes* infection.

Cont.

- In acute infections, *L. Monocytogenes* evokes an exudative pattern of inflammation with numerous neutrophils.
- The finding of gram-positive, mostly intracellular bacilli in the CSF is virtually diagnostic.

Anthrax

- Anthrax is characterized by necrotizing inflammatory lesions in the skin or gastrointestinal tract or systemically.
- It is caused by *Bacillus anthracis*, a large, spore forming gram-positive rod-shaped bacterium found in environmental sources.

Cont

- There are three major forms of anthrax.
- Cutaneous anthrax, which makes up 95% of naturally occurring infections, begins as a painless, pruritic papule that develops into a vesicle within 2 days.

Cont.

- As the vesicle enlarges, striking edema may occur around it, with development of regional lymphadenopathy.
- After the vesicle ruptures, the remaining ulcer becomes covered with a characteristic black eschar, which dries and falls off as the person recovers. Bacteremia is rare.

Cont

- Inhalational anthrax occurs when airborne spores are inhaled. The spores are carried by phagocytes to lymph nodes where they germinate, producing bacilli that release toxins that cause hemorrhagic mediastinitis.
- Inhalational anthrax rapidly leads to shock and frequently death within 1 to 2 days.

Cont

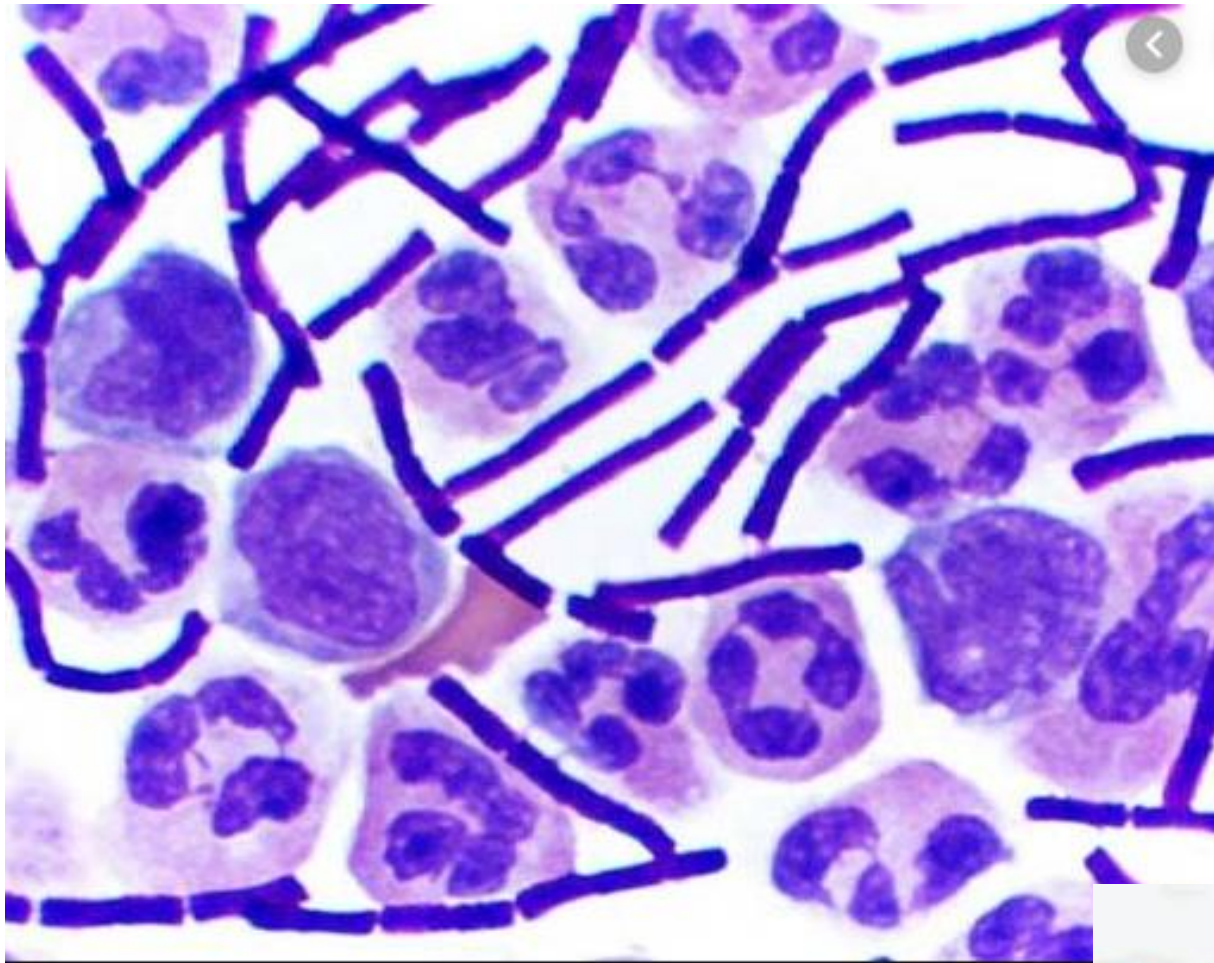
- Gastrointestinal anthrax is usually contracted by eating undercooked meat contaminated with *B. anthracis*.
- Initially, the person has nausea, abdominal pain, and vomiting, followed by severe, bloody diarrhea and, sometimes, bacteremia.
- Mortality is approximately 40%.

Pathogenesis

- *B. anthracis* produces potent toxins and an antiphagocytic polyglutamyl capsule.

Microscopic

- Anthrax lesions at any site are typified by necrosis and exudative inflammation rich in neutrophils and macrophages.
- The presence of large, boxcar-shaped gram-positive extracellular bacteria in chains, seen histopathological or grown in culture, suggests the diagnosis.
- The lungs typically show a perihilar interstitial pneumonia with infiltration of macrophages and neutrophils and pulmonary vasculitis.



Nocardia

- Nocardia are aerobic gram-positive bacteria found in soil that cause opportunistic infections.
- The organism grows in distinctive branched chains. In culture, Nocardia form thin aerial filaments resembling hyphae.

Cont.

- Nocardia causes respiratory infections, most often in patients with defects in immunity.
- Respiratory infection with *N. asteroides* causes an indolent illness with fever, weight loss, and cough, which may be mistaken for tuberculosis or malignancy.

Cont

- Nocardia appear in tissue as slender gram-positive organisms arranged in branching filament
- Irregular staining gives the filaments a beaded appearance. Nocardia stain with **modified acid-fast stains** (Fite-Faraco stain).
- Nocardia elicit a suppurative response with central liquefaction and surrounding granulation and fibrosis.
- Granulomas do not form.

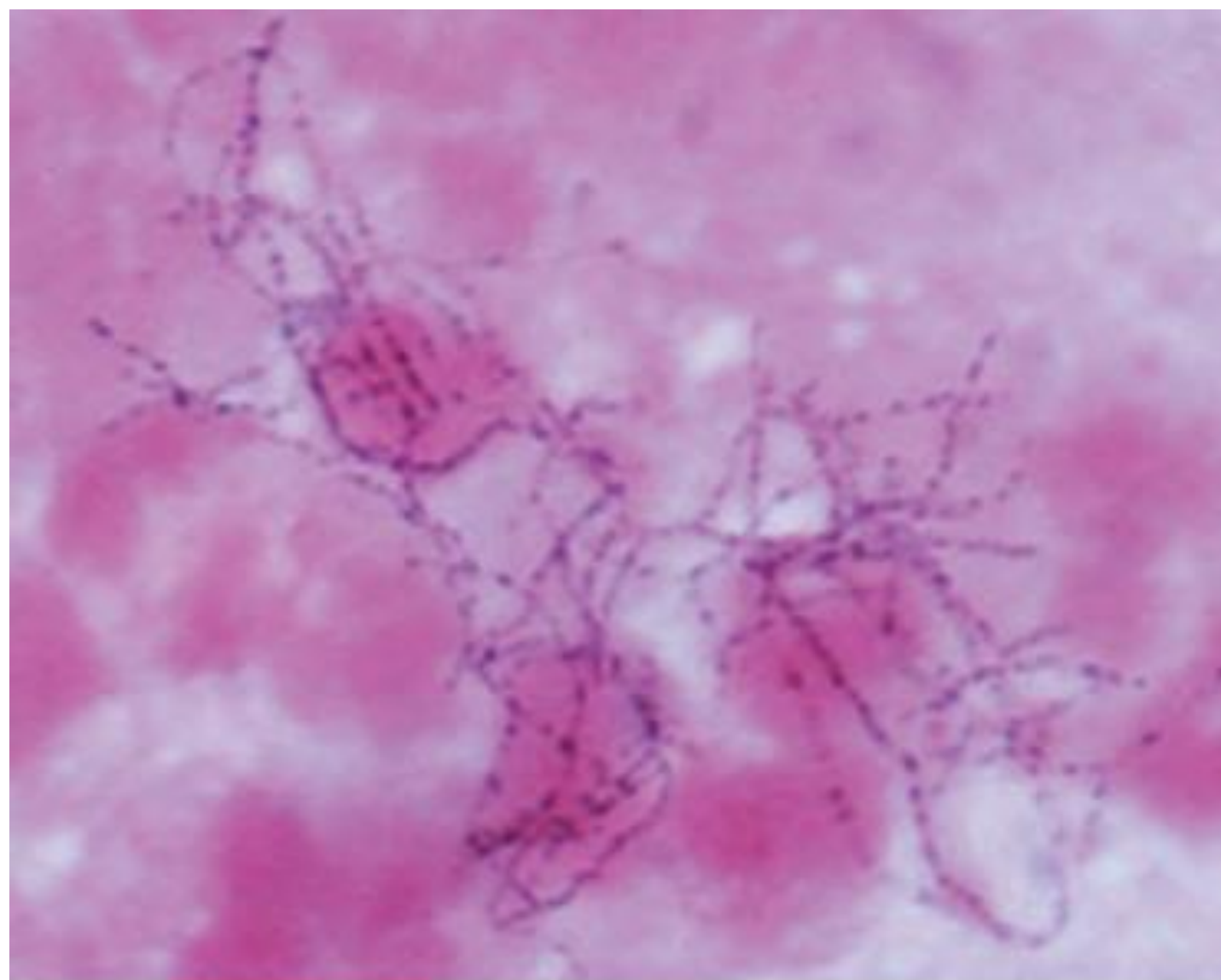


Figure 8-21 *Nocardia asteroides* in a Gram-stained sputum smear showing the beaded, branched gram-positive organisms and leukocytes.

Gram negative bacteria

Neisserial Infections

- Neisseria are gram-negative diplococci that are flattened on the adjoining sides, giving the pair the shape of a **coffee bean**.
- The two clinically significant Neisseria are N. meningitidis and N. gonorrhea.
- N. meningitidis is a significant cause of bacterial meningitis, particularly among adolescents and young adults.

Cont.

- *N. gonorrhoeae* is an important cause of sexually transmitted disease (STD), Infection in men causes urethritis.
- In women, *N. gonorrhoeae* infection is often asymptomatic and so may go unnoticed.
- Untreated gonorrhea may lead to pelvic inflammatory disease, which can cause infertility or ectopic pregnancy.
- Infection is diagnosed by culture and PCR tests.

Pertussis

- Pertussis, or whooping cough, caused by the gram negative coccobacillus *Bordetella Pertussis*.
- highly communicable illness characterized by paroxysms of violent coughing followed by a loud inspiratory “whoop.”
- The diagnosis is best made by PCR, because culture is less sensitive.

Cont

- Bordetella bacteria cause a laryngotracheobronchitis that in severe cases features bronchial mucosal erosion, hyperemia, and copious mucopurulent exudate.

Cont.

- In parallel with a striking peripheral lymphocytosis (up to 90%), there is hypercellularity and enlargement of the mucosal lymph follicles and peribronchial lymph nodes.

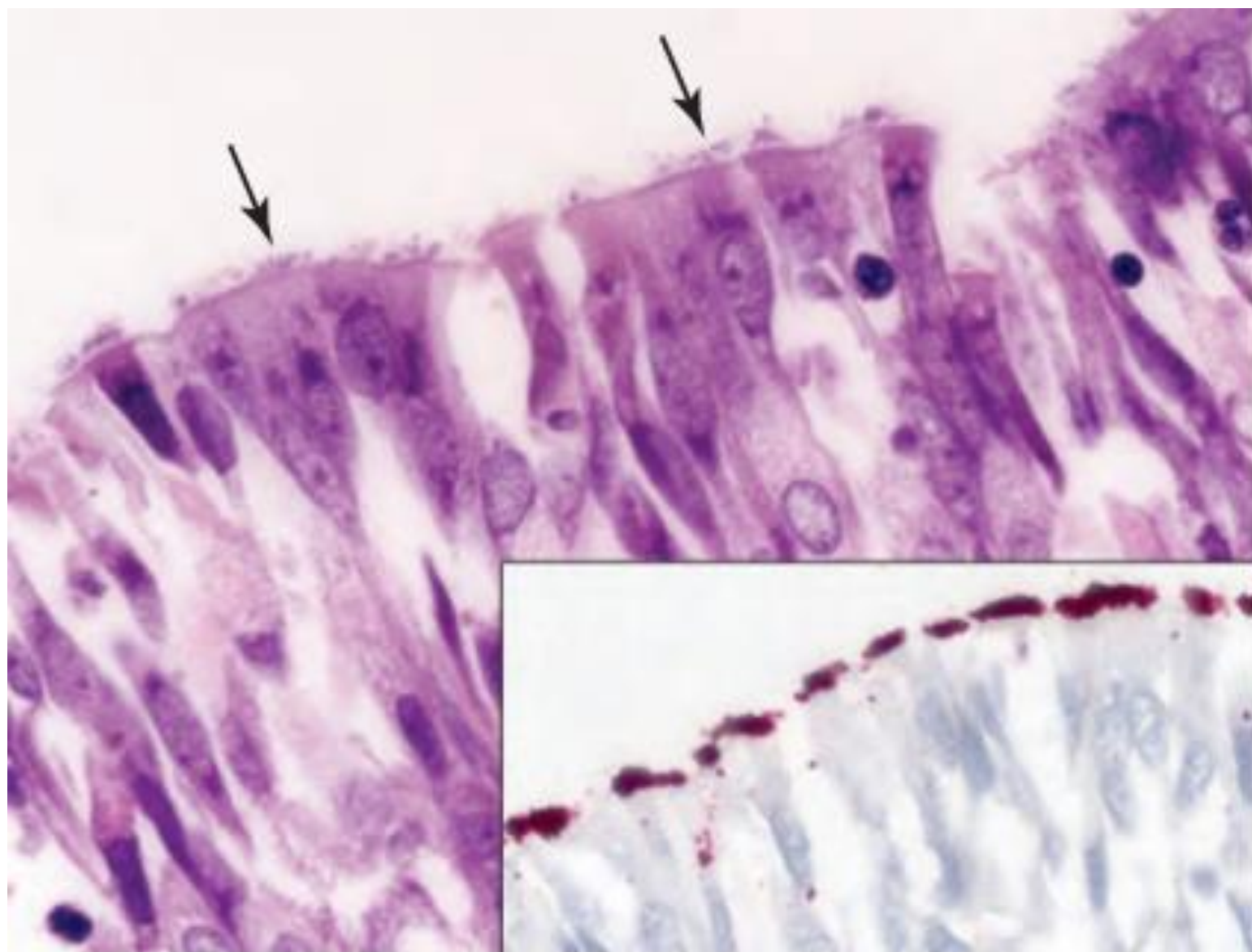


Figure 8-22 Whooping cough showing a haze of bacilli (*arrows*) entangled with the cilia of bronchial epithelial cells. The inset highlights the haze of bacilli

Pseudomonas Infection

- *Pseudomonas aeruginosa* is an opportunistic aerobic gram-negative bacillus that is a frequent, deadly pathogen of people with cystic fibrosis, severe burns, or neutropenia.
- Many people with cystic fibrosis die of pulmonary failure secondary to chronic infection with *P. aeruginosa*.
- *P. aeruginosa* can be very resistant to antibiotics, making these infections difficult to treat.

Cont

- It also causes corneal keratitis in wearers of contact lenses, endocarditis and osteomyelitis in intravenous drug abusers, external otitis (swimmer's ear) in healthy individuals, and severe external otitis in people with diabetes.

cont

- Pseudomonas causes a necrotizing pneumonia that is distributed through the terminal airways.

Cont.

- On microscopic examination, masses of organisms are seen that tend to be most concentrated in the walls of blood vessels, where host cells undergo coagulative necrosis with picture of gram-negative bacterial vasculitis.

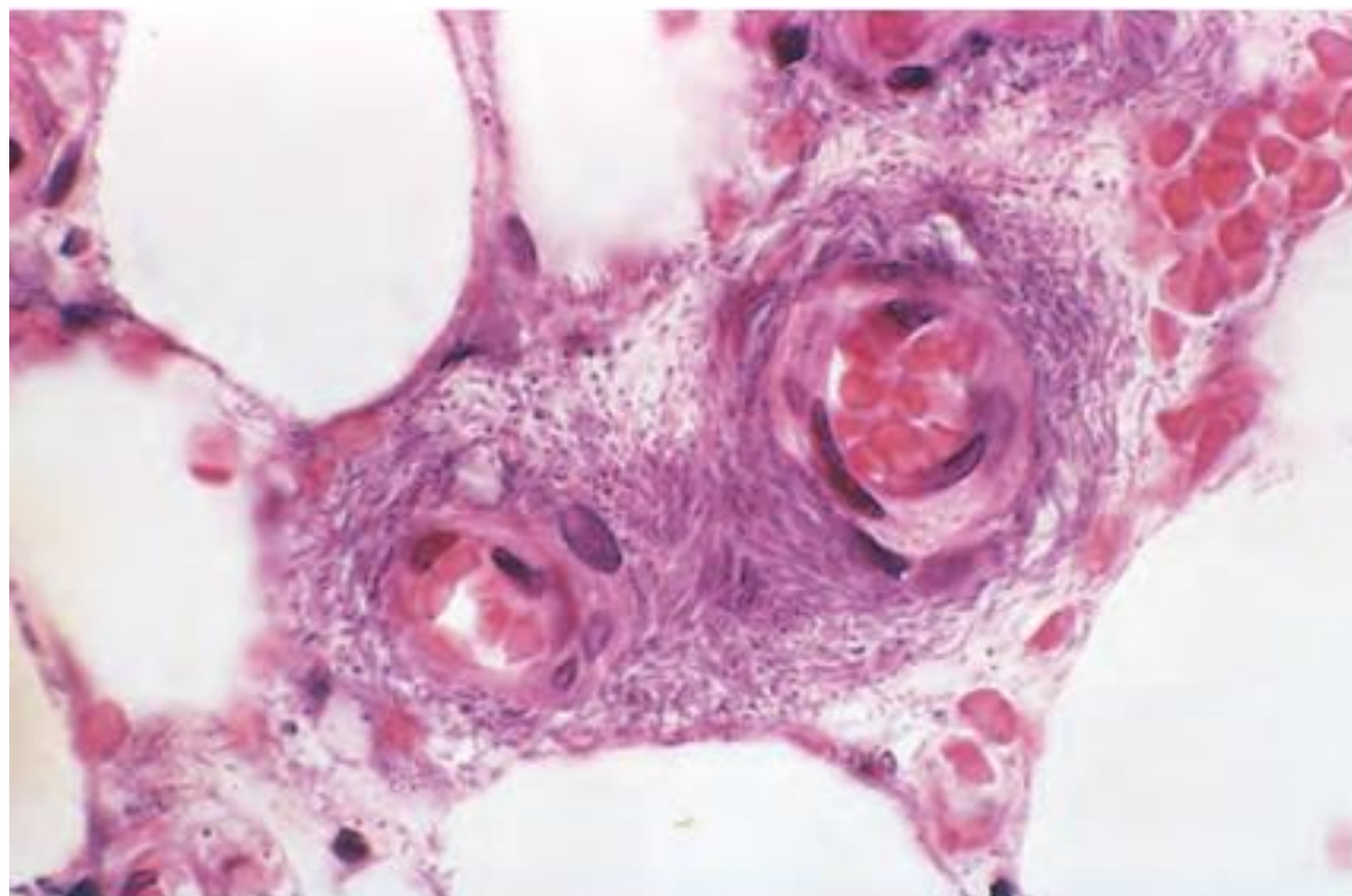


Figure 8-23 *Pseudomonas* vasculitis in which masses of organisms form a perivascular blue haze.

Plague

- *Yersinia pestis* is a gram-negative facultative intracellular bacterium that causes an invasive, frequently fatal, infection called plague.
- It is transmitted from rodents to humans by fleabites or, less often, from one human to another by aerosols.

Cont.

- Plague, also named Black Death, caused three great pandemics that killed an estimated 100 million people in Egypt and Byzantium in the sixth century; one quarter of Europe's population in the fourteenth and fifteenth centuries; and tens of millions in India, Myanmar, and China at the beginning of the twentieth century.

Cont

- *Yersinia pestis* causes lymph node enlargement (buboes), pneumonia, or sepsis with a striking neutrophilia.
- In pneumonic plague there is a severe, confluent, hemorrhagic and necrotizing bronchopneumonia, often with fibrinous pleuritis

Chancroid (Soft Chancre)

- Chancroid is an acute, sexually transmitted, ulcerative infection caused by *Haemophilus ducreyi*.
- The disease is most common in tropical and subtropical areas among lower socioeconomic groups and men who have frequent sex with prostitutes.

Cont

- The organism is difficult to grow in culture and PCR-based tests are not widely available, so Chancroid is likely to be under diagnosed.

Cont.

- Microscopically, the ulcer of Chancroid contains a superficial zone of neutrophilic debris and fibrin, and an underlying zone of granulation tissue containing areas of necrosis and thrombosed vessels.

Cont

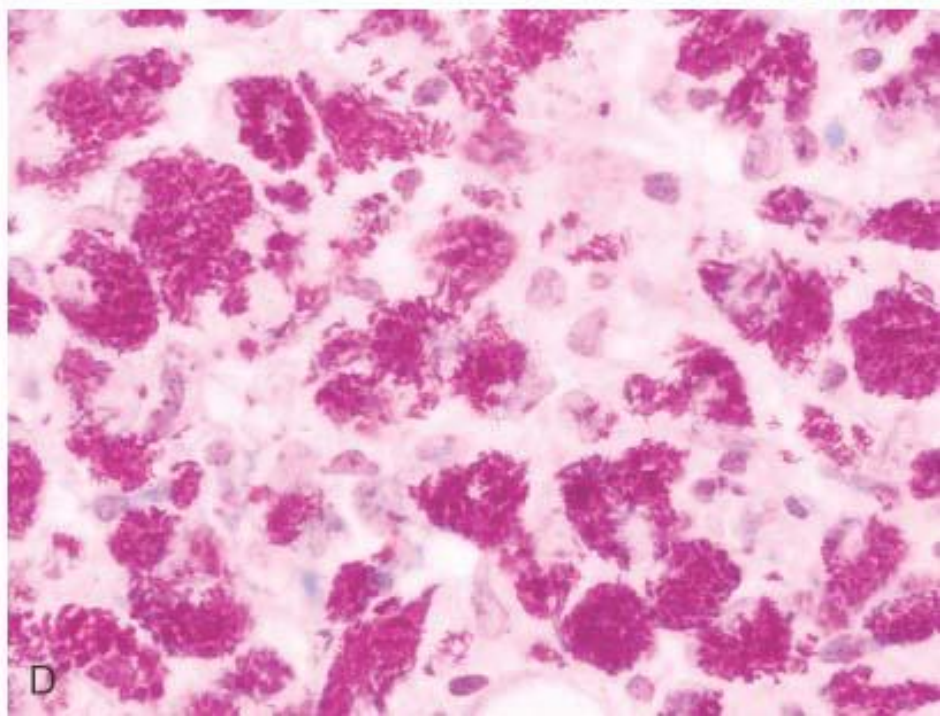
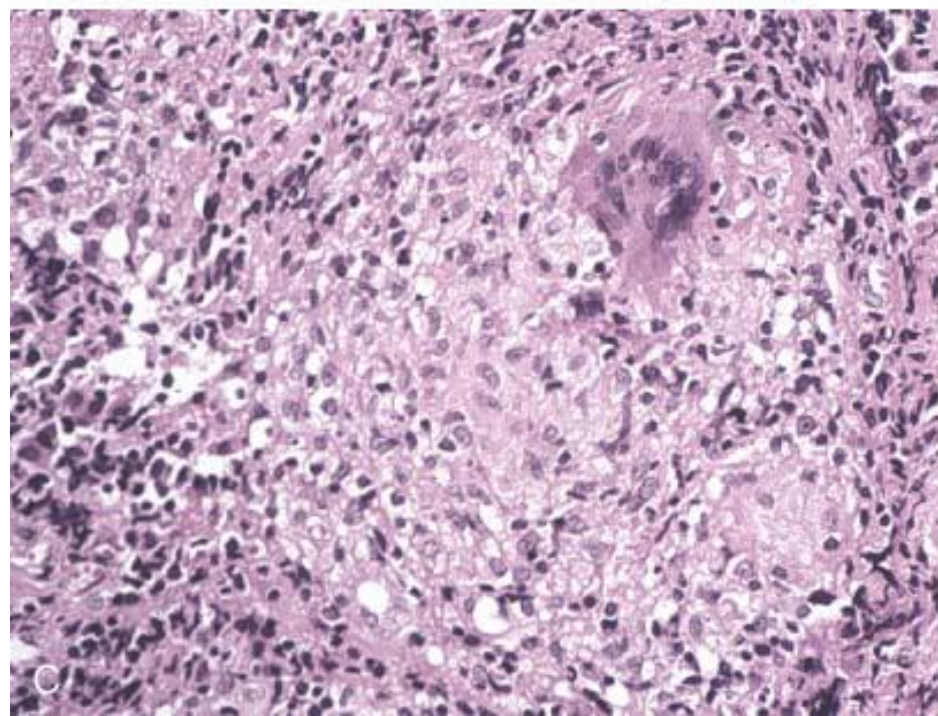
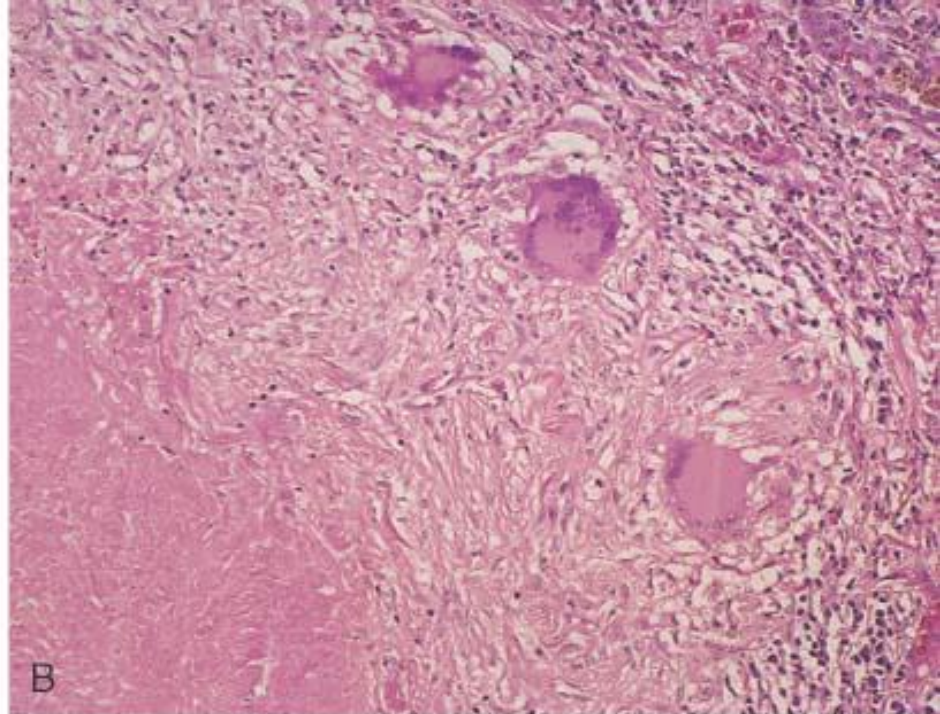
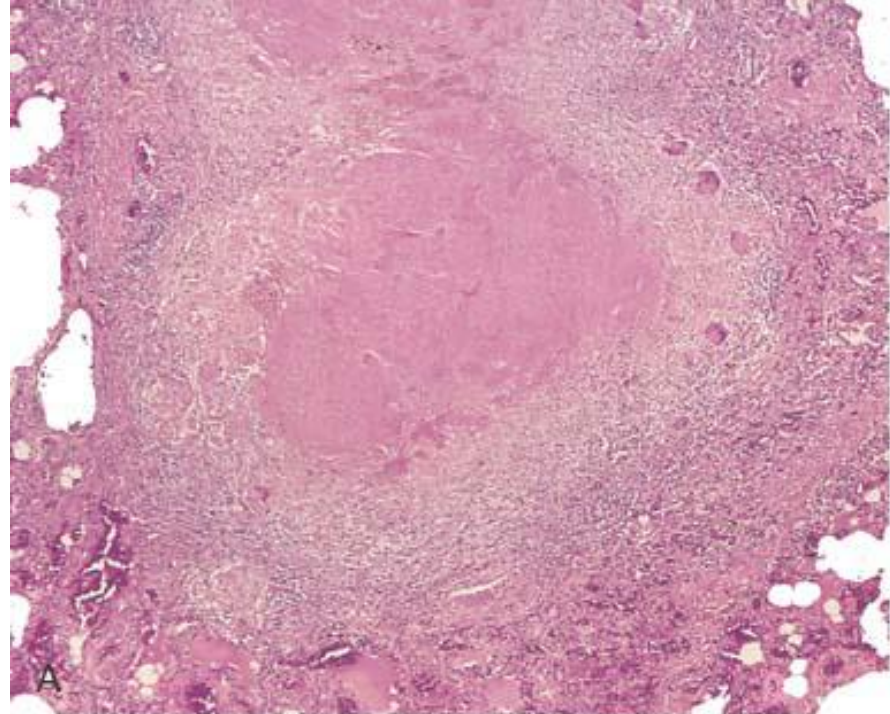
- A dense, lympho plasmacytic inflammatory infiltrate is present beneath the layer of granulation tissue.
- **Coccobacilli** are sometimes demonstrable in Gram or silver stains.

Mycobacterium tuberculosis

- Clinical manifestations
- Diagnosis

Cont

- Histologically, sites of active disease are marked by granulomatous inflammatory reaction that forms caseating & non caseating granuloma.
- The granuloma are enclosed within fibroblastic rim & cuff of lymphocytes.
- Immunocompromised people don't form the characteristic granuloma.



Mycobacterium Avium –Intracellulare complex

- Mycobacterium avium & Mycobacterium intercellulare are separate species but are referred as *Mycobacterium avium-intercellulare complex (MAC)*
- MAC is common in soil, water, dust & domestic animals .
- Clinically significant infection with MAC is uncommon except among people with AIDS & low CD4 lymphocyte($<60/\text{mm}^3$)

Cont

- In AIDS patients , MAC cause widely disseminated infection & organisms proliferate abundantly in many organs commonly in the lungs & gastrointestinal system.

Morphology

- The hallmark of MAC infection in patients with HIV is abundant acid-fast bacilli within macrophages

cont



